USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Clarifications for LFPS_TEST Port Operation
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Changed the description of the LFPS_TEST Port Operation so it will indicate that both RX and TX might be tested and add a statement about which transmitter is used when testing which receiver.

Benefits as a result of the changes:

A full description of the Port Operation. Allowing testing of RX2 when Port support Asymmetric Link with 3 receivers.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:

None

An analysis of the hardware implications:

None

An analysis of the software implications:

None

An analysis of the compliance testing implications:

Electrical CTS will have a dedicated test for receivers LFPS mechanis.

Actual Change

(a). Table 8-76 – LFPS_TEST Operation Metadata

To Text:

DW	Bit(s)	Field Name and Description
0	5:0	**Port** – Identifies the target USB4 Port. For a Router: This field contains the Adapter Number of the Lane 0 Adapter of the target USB4 Port. For a Re-timer: 0h: Target is the USB4 Port whose SB Register Space is written to. 1h: Target is the USB4 Port whose SB Register Space is not written to.
	8:6	**Adapter** – Identifies the Adapter within the USB4 Port. 000b: TX0 (for Modes 000b and 001b)\RX0 (for Modes 010b and 011b) 001b: TX1 (for Modes 000b and 001b)\RX1 (for Modes 010b and 011b) 010b: TX2 (for Modes 000b and 001b)\RX2 (for Modes 010b and 011b) (only applies to a Port that supports Asymmetric Link with 3 Tx or 3 Rx) All other values are reserved.
	11:9	**Mode –** This field defines the LFPS test mode. 000b: Adapter shall transmit LFPS continuously. 001b: Adapter shall transmit 100 cycles of LFPS, transition to Electrical Idle for tPreData and start sending SLOS1 (for Gen 2 or Gen 3) or PRBS11 (for Gen 4). 010b: Adapter shall transmit LFPS continuously after detecting at least 3 LFPS cycles on the selected~~its~~ receiver. 011b: Adapter shall transmit LFPS when it detects LFPS on ~~its~~the selected receiver. Adapter shall move to Electrical Idle when not detecting LFPS on ~~its~~the selected receiver. Note: In Modes 010b and 011b, when Link is Symmetric, the transmission of LFPS will be on selected Lane transmitter, When the Link is Asymmetric, the transmission of LFPS will be on TX0.
	31:12	**Reserved**